Randomized and low-rank approximation

Sensitivity-dependent row sampling for ℓ_p embeddings when p exceeds two

Difficulty: challenging **Importance:** interesting to the community

Status: Lean verified

Rating rationale: Challenging because ordinary sensitivities must control all directions with nearly linear total-sensitivity cost; community impact is instance-sensitive regression and subspace sketching.

Statement

Fix $p > 2$. For a full-column-rank matrix $A \in \mathbb{R}^{n\times d}$, with rows a_i^T, set

$$s_i = \sup_{x\neq 0} \frac{|a_i^T x|^p}{\|Ax\|_p^p}, \qquad \mathcal{S} = \sum_{i=1}^n s_i.$$

For a scalar $\alpha > 0$, retain row i independently with probability

$$q_i = \min\{1, 1/n + s_i/\alpha\},$$

and scale each retained row by $q_i^{-1/p}$; let $\widetilde{A}$ be the resulting matrix.

Do constants $C_p, c_p > 0$ exist such that, for every A and $0 < \varepsilon, \delta < 1/2$, one can choose α with

$$\sum_i q_i \le C_p \varepsilon^{-2}(\mathcal{S} + d)\log^{c_p}\left(\frac{2nd}{\varepsilon\delta}\right),$$

while, with probability at least $1 - \delta$,

$$(1-\varepsilon)\|Ax\|_p^p \le \|\widetilde{A}x\|_p^p \le (1+\varepsilon)\|Ax\|_p^p \qquad \text{for all } x \in \mathbb{R}^d?$$

The size is the expected number of retained rows. The question concerns this independent sampling rule based on the ordinary ℓ_p sensitivities, not arbitrary row weights or augmented sensitivities.

This would exploit easy instances through total sensitivity rather than using the worst-case dimension-only bound. It applies directly to sketching ℓ_p regression and other computations on the column space of A. The p>2 ordinary-sensitivity theorem of Woodruff–Yasuda has size $\widetilde{O}_p(\varepsilon^{-2}\mathcal{S}^{2-2/p})$.

Resolution — 2026-09-30

Negative resolution for every fixed real exponent $p > 2$. Theorem 5.1 of *An Accuracy Obstruction for lp Row Sparsification: A Complete-Graph Counterexample to RA-06* (LaTeX source), dated 30 September 2026, refutes the original expected-size guarantee with exactly its ordinary sensitivities, independent retention rule, rescaling and simultaneous norm-power preservation.

The grounded incidence matrix of the complete graph on v vertices has $n = \binom{v}{2}$, $d = v - 1$, and equal row sensitivities

$$s_i = \frac{1}{1 + (v-2)2^{1-p}}, \qquad \mathcal{S} = \Theta_p(d).$$

Under $v - 1 \ge 6^{-p}\varepsilon^{-(p+1)}$, Corollary 4.2 forces expected sample size $\Omega_p((\mathcal{S} + d)\varepsilon^{-p})$ even for one successful realization of the prescribed common-probability sampler. Taking $\varepsilon = b^{-1}$, $\delta = 1/4$ and $v =$

$\lceil b^{p+2} \rceil$ contradicts every fixed polylogarithmic multiplier of the proposed quadratic inverse-accuracy budget. The floor and saturation in the original rule are included.

Theorem 3.1 and Corollary 3.2 also give a support obstruction for arbitrary nonnegative reweighted row subsets, with a success-probability factor for randomized expected support. The obstruction concerns accuracy dependence in an explicit growing-dimension regime; it does not rule out nearly linear sensitivity dependence at fixed accuracy and does not concern unrestricted linear sketches.

Attribution and review. The original conjecture and sampling model are credited to Woodruff–Yasuda, Section 3 and Definition 1.1/Theorem 1.5. **No novelty or priority is claimed for this local manuscript.** The supplied ChatGPT 6 Pro audit reports that the mathematical core refutes RA-06 and derives the arbitrary-weight strengthening. Its follow-up revision incorporates that strengthening, direct attribution, the explicit dimension regime and the requested presentation fixes. This manuscript review is an informal AI audit, not external human peer review; the later formal checks are in the evidence section. See the submission record. Historical ratings are retained, and the original target is retained above.

Lean proof and verification evidence

Reviewed 13 October 2026.

At immutable repository revision *bca7904b*, the proof defines *NLA.Proofs.RA06.target*. Its literal type is *NLA.Statements.RA06.Target*: for every real exponent greater than two, it refutes the original positive claim. The independent mathematical and exact-statement review checks the unchanged ordinary sensitivities, independent retention probabilities and rescaling, one common sampling parameter, expected row count, and simultaneous embedding quantifiers. The proof adds no premise and does not substitute a different sampler.

The source-locked project pins Lean **4.33.1**, Mathlib *0df444a360ea*, and LeanCert *621a43d7cf21*. From *randomized-and-low-rank-approximation/RA-06/lean*, run *lake build Challenge Solution*; from *lean-statements*, run *lake build NLA.Proofs.RA06.Final*. The 10 October isolated Linux Comparator/LeanCert job and log passed for the selected *NLA.RA06.target* export. Its transitive axiom report contains only *propext*, *Classical.choice*, and *Quot.sound*; the receipt records the successful revision and reproduction details. This catalog reviewed that public record and the unchanged reviewed source hash; it did not rerun Lean locally for this status update.

References

1. D. P. Woodruff and T. Yasuda, *Sharper Bounds for ℓ_p Sensitivity Sampling*. ICML 2023, arXiv:2306.00732v2. Definition 1.1 fixes the sampling model, Theorem 1.5 gives the p>2 bound, and Section 3 explicitly conjectures $\widetilde{O}(\varepsilon^{-2}(\mathcal{S}+d))$. Paper.
2. A. Munteanu and S. Omlor, *Optimal bounds for ℓ_p sensitivity sampling via ℓ_2 augmentation*. ICML 2024, PMLR 235, pp. 36769–36796. Question 1.2 and Section 1.1 distinguish the solved/obstructed p≤2 situation from the remaining p>2 case. Paper.
3. R. Chhaya, A. Dasgupta, D. Feldman, and S. Shit, *Deterministic Coreset for Lp Subspace*. arXiv:2601.00361 (2026). This claimed deterministic embedding result was withdrawn on May 15, 2026 because some proofs are incomplete; it is cited only to prevent a stale abstract from being mistaken for a resolution. Paper.

Status check — 2026-09-08

Searched "p>2 sensitivity sampling conjecture 2025 2026", "sensitivity sampling S+d", and the exact paper titles; checked the current arXiv records. No resolution of the stated p>2 rule was found. Results using ℓ_2 augmentation for p≤2 do not settle it. The 2026 deterministic embedding claim was withdrawn (latest v3, May 15, 2026), so its abstract is not evidence that any open case has been resolved.

Audit — 2026-09-10

Rechecked Woodruff–Yasuda, §3 and the augmentation paper's scope. The ordinary-sensitivity $p > 2$ target remains unresolved in these sources. Sensitivity-sampling follow-up searches found no resolution. Difficulty was raised to reflect the general uniform sampling obstruction.